

MATHEMATICS FOR PROBLEM SOLVING

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COURSE OUTCOMES:

- C01:** Understand essential concepts of time and work, including work efficiency and wage distribution for solving practical problems.
- C02:** Enhance problem-solving and analytical skills through permutations, combinations, and ranking-based questions.
- C03:** Develop a strong foundation in probability theory and apply it to real-life scenarios involving coins, dice, and cards.
- C04:** Master the concepts of clocks and calendars, including angles, time loss/gain, and calendar repetitions for effective time management.
- C05:** Learn to interpret and analyze data from tables, graphs, and charts to make informed decisions and solve real-world problems.
- C06:** Strengthen logical thinking and pattern analysis skills through ranking, direction, and coding-decoding problems to improve decision-making.

MODULE I:

TIME & WORK

Time and Work: Work efficiency concepts, pipes and cisterns, wage distribution, and the man- chain rule for determining work completion timelines.

MODULE II:

PnC & PROBABILITY

Permutations & Combinations: Fundamental counting principle, permutation and combination formulas, ranking-based questions, and circular permutations. Probability: Solving problems related to probability with coins, dice, playing cards, and miscellaneous probability-based scenarios.

MODULE III:

CLOCKS AND CALENDARS

Clocks: Angle between hands, gaining and losing time, problems on incorrect clocks, Calendars: Odd days concept, leap year calculations, Calendar Repetition.

MODULE IV:**DATA INTERPRETATION AND ITS APPLICATIONS**

Tables and Graphs: Interpretation of data presented in tables, bar graphs, pie charts, and line graphs. Applications: Analyzing trends, calculating percentages, ratios, and averages, and comparing data across different categories.

MODULE V:**LOGICAL THINKING AND PATTERN ANALYSIS**

Ranking: Problems with relative and absolute ranking, finding a position in a row or queue, and highest and lowest ranking scenarios. Directions: Cardinal directions (North, South, East, West), distance calculation, final direction and position determination, clockwise and counterclockwise movement. Coding-Decoding: Letter shifting, number coding, symbol coding, word-pattern coding, substitution, and deciphering logical sequences.

TEXT BOOKS:

1. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal
2. "Problems in Probability, Statistics and Random Processes" by T. Veerarajan
3. "Analytical & Logical Reasoning" by B.S. Sijwali & Indu Sijwali

FUNDAMENTALS OF QUANTUM COMPUTING

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Course Outcomes:

CO1: To be exposed to the fundamental concepts of Wave nature of Particles and Particle nature of Waves.

CO2: Understand the formation of energy bands in solids

CO3: Apply Quantum mechanics in the emerging field of Quantum computing.

CO4: Students will be able to interpret and construct basic quantum circuit diagrams.

CO5: Students will be able to identify and discuss real-world applications of quantum computing

MODULE - I

QUANTUM MECHANICS: Introduction to quantum physics, blackbody radiation, photoelectric effect-Broglie hypothesis, Davisson and Germer experiment, Heisenberg uncertainty principle, Born interpretation of the wave function, time dependent and independent Schrodinger wave equation - particle in one dimensional potential box.

MODULE - II

BAND THEORY OF SOLIDS

Classical and Quantum Free electron theory assumptions, Merits and drawbacks (qualitative), Bloch theorem, Kronig- Penny model (qualitative), E-k diagram, Origin of energy bands in solids: Classification of Solids - Conductors, Semiconductors and Insulators.

MODULE - III

INTRODUCTION QUANTUM COMPUTING

Introduction, Classical & quantum bits, Superposition and Entanglement, Qubits Operators and Measurement: Bra-Ket (Dirac) notation, Bloch sphere representation of a Qubit, Classical logic gates, Quantum logic gates: Single qubit gates (Pauli-X, Pauli-Y, Pauli-Z, Hadamard), multiple qubit gates (CNOT gate)

MODULE - IV

QUANTUM CIRCUITS

Introduction to Quantum Circuits, Differences between classical and quantum circuits, Basic Circuit Notation, Quantum circuit diagrams, superposition, and entanglement using quantum circuits.

MODULE - V

BRIEF APPLICATIONS OF QUANTUM COMPUTING

Molecular modelling and design of new materials, deep learning, cyber security, quantum neural networks and artificial intelligence. Ethical considerations in quantum computing.

REFERENCE BOOKS:

1. Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci
2. Chris Bernhardt, Quantum Computing for Everyone, MIT Press, Illustrated Edition, April 2019.
3. Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II
4. Basic Tools and Special Topics, World Scientific. Pittenger A. O., an Introduction to Quantum Computing Algorithms
5. David McMahon, Quantum computing explained, Wiley-interscience, John Wiley & Sons, Inc. Publication 2008.
6. Quantum Computing in Practice with Qiskit and IBM Quantum Experience, Hassi Norlén, 2020.

TEXT BOOKS:

1. Michael A. Nielsen and Isaac L. Chuang, Quantum computation and quantum information, Cambridge University Press, 2010.

DATA STRUCTURES

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Course Outcomes:

- CO1:** Understand the foundational concepts of asymptotic analysis and time-space complexity.
- CO2:** Develop proficiency in recursive algorithms and their applications in solving mathematical and computational problems.
- CO3:** Gain expertise in linear data structures such as arrays, strings, linked lists, stacks, and queues, and apply them to solve real-world challenges.
- CO4:** Master non-linear data structures, including trees, heaps, and graphs, along with their traversals and applications in shortest paths, spanning trees, and graph connectivity.
- CO5:** Implement sorting and searching techniques, both basic and advanced, to optimize problem-solving.
- CO6:** Utilize advanced data structures such as hashing, tries, and disjoint sets for efficient data organization and retrieval.

MODULE - I

BASIC DATA STRUCTURES AND COMPLEXITY ANALYSIS:

Introduction, Basic Terminology, Data Structures, Algorithms, Asymptotic Notations, Time and Space Complexity Analysis, Linear Arrays, Representation of Linear Arrays in Memory, Traversing Linear Arrays, Searching, Advanced Array Applications: Prefix Sum, Sliding Window, Two-Pointer Techniques, Multidimensional Arrays, Pointers, Recursion Fundamentals, Recursive Algorithms and Applications, Towers of Hanoi

MODULE - II

SORTING AND SEARCHING TECHNIQUES

Basic Sorting: Selection Sort, Bubble Sort, Insertion Sort, Advanced Sorting: Merge Sort, Quick Sort, Binary Search and Applications, Searching and Data Modification, Hashing, Hash Tables and Hash Functions.

MODULE - III

LINEAR DATA STRUCTURES

Linked List, Representation of Linked List in Memory, Traversing a Linked List, Searching a Linked List, Memory Allocation, Insertion into a Linked List, Deletion from a Linked List, Header Linked Lists, Two-way Lists, Advanced Operations: Reversals, Merging, Flattening, Strings, String Manipulation Techniques.

MODULE - IV STACKS, QUEUES

Stacks, Array Representation of Stacks, Arithmetic Expressions: Infix, Prefix, Postfix, Applications of Stacks, Monotonic Stack and Next Greater Element, Implementation of Recursive Procedures by Stacks, Queues, Implementing Stack using Queue, Implementing Queue using Stack, Linked Representation of Queues, Dequeues, Priority Queues.

MODULE - V

NON-LINEAR DATA STRUCTURES

Binary Trees, Binary Tree Traversals, Applications: Path Sum Calculations, Lowest Common Ancestor, Binary Search Trees, Heaps: Min-Heap, Max-Heap, Priority Queue Implementation, Heap Sort, Huffman's Algorithm, Graph Theory Terminology, Graph Representation, Graph Traversals techniques, BFS, DFS, Cycle Detection, Topological Sorting, Bipartite Graph

LIST OF EXPERIMENTS:

Design & implement the following concepts:

1. Linked List
2. Stack ADT
3. Queue ADT
4. Priority Queue
5. Sorting and Searching
6. Tree Traversal
7. Tree Structure - Binary Tree
8. Binary Tree Structure
9. Graph Traversals (BFS, DFS)
10. Hash Table

REFERENCE BOOKS:

1. Fundamentals of Data Structures in C++-By Sartaj Sahani.
2. Data Structures: A Pseudo-code approach with C -By Gilberg & Forouzan Publisher-Thomson Learning.

TEXT BOOKS:

1. An Introduction to Data Structures with Applications. by Jean-Paul Tremblay & Paul G Sorenson Publisher-Tata McGraw Hill.
2. Data Structures using C & C++ -By Ten Baum Publisher – Prentice-Hall International
3. Fundamentals of Computer Algorithms by Horowitz, Sahni, Galgotia Pub. 2001 ed.

DATABASE MANAGEMENT SYSTEMS

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Course Outcomes:

C01: Understand the principles of relational databases, including database design, ACID properties, and their applications.

C02: Perform essential SQL operations such as creating, inserting, selecting, updating, and deleting records in a database.

C03: Utilize advanced SQL techniques, including joins, subqueries, indexes, and views, to optimize database performance.

C04: Apply database normalization techniques to organize data efficiently, eliminate redundancy, and ensure integrity.

C05: Execute advanced database operations such as truncate, drop, and create stored procedures and triggers, while gaining practical skills to design, manage, and query databases for real-world applications.

C06: Develop hands-on proficiency with NoSQL databases using MongoDB, including creating collections, performing CRUD operations, querying

MODULE I:

INTRODUCTION TO DATABASE SYSTEMS

Database Fundamentals, DBMS Types, Relational vs Non-Relational Databases, Database Approach Benefits, Data Models, Schemas and Instances, DBMS Components, Database Architecture, Entity Relationship Model, ER Diagram Notation, ER Design Refinement, ER to Relational Mapping, Relationship Types, Implementation Strategies.

MODULE II:

SQL FUNDAMENTALS

Constraints and Schemas, Unary Operations: SELECT, Set Theory Operations, SQL Basics, Table Creation and Modification, Data Manipulation, WHERE Clause, Primary and Foreign Keys.

MODULE III:

ADVANCED SQL AND QUERY OPERATIONS

Comparison and Logical Operators, Pattern Matching with LIKE, Operator Precedence, IN and BETWEEN Operators, Result Ordering, LIMIT and OFFSET, Set Operations, SQL Functions, Triggers, CASE Expressions, Expression Usage in Queries, Multi-table Queries, Join Types, Aggregation Functions, Alias Usage, GROUP BY Implementation, HAVING Usage, View Creation and Management, Subqueries, Query Processing and Optimization.

MODULE IV:

DATABASE NORMALIZATION AND TRANSACTION PROCESSING

Schema Refinement, Functional Dependencies, Normalization: Normalization -1 NF - 2 NF - 3 NF - Boyce-Codd Normal Form, Multi-valued Dependencies and Fourth Normal Form, Join Dependencies and Fifth Normal Form, Domain Key Normal Form, Denormalization, ACID Properties, Transaction Management, Concurrency Control.

MODULE V:

INTRODUCTION TO NOSQL DATABASES AND MONGODB

NoSQL Database Concepts, Collections vs Tables, Documents vs Records, Fields, Schema Flexibility, Setting up MongoDB and Creating a Database, CRUD Operations in MongoDB, Querying: Projection, Conditional Operators, Ordering Results, Pagination, Logical Operators, Nested Documents, Array Fields, Basic Array Queries, Array Operators, \$elemMatch, Grouping & Aggregate functions, Comparison of SQL vs NoSQL

PRACTICALS:

- Creating, altering and dropping tables with integrity constraints using DDL commands.
- Retrieving and modifying data from a database using DML Statements.
- Retrieving data from the database using IN, BETWEEN, LIKE, ORDER BY, GROUP BY and HAVING clauses.
- Use of scalar and aggregate functions.
- Retrieving data from a database using Join operations (Inner and Outer Joins).
- Using sub queries for retrieving data.
- Use of trigger in data manipulation
- Design and Develop a NoSQL Database

REFERENCE BOOKS:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition.
2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education
3. Introduction to Database Systems, C. J. Date, Pearson Education
4. Oracle for Professionals, The X Team, S.Shah and V. Shah, SPD.
5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI.
6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition.

TEXT BOOKS:

1. Database System Concepts, Silberschatz, Korth, McGraw hill, V edition.3rd Edition
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill

WEB APPLICATION DEVELOPMENT-II

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Course Outcomes:

- CO1:** Master modern JavaScript concepts including scope, hoisting, and ES6 features for building interactive web applications.
- CO2:** Implement asynchronous programming techniques using callbacks, promises, and async/await to create responsive frontend applications.
- CO3:** Build functional user interfaces with React, leveraging components, props, state, and lifecycle methods.
- CO4:** Develop complex React applications using hooks, routing, and global state management solutions.
- CO5:** Create secure frontend applications with proper authentication flows and API integration.
- CO6:** Apply industry best practices for frontend architecture, performance optimization, and code maintainability.

MODULE I:

MODERN JAVASCRIPT

Scope: Global Scope, Function Scope, Block Scope. Hoisting: Variable Hoisting, Function Hoisting. 'this' Keyword and Context. Error Handling: try/catch/finally. ES6 Features: JavaScript Classes, Arrow Functions, Lexical Scope, Template Literals, Destructuring Syntax, Spread Syntax. Array Methods. String Methods. Object Methods (keys, values, entries).

MODULE II:

ASYNCHRONOUS JAVASCRIPT

Asynchronous Programming Basics. Callback Functions. Callback Hell and Management. Timer Functions (setTimeout, setInterval). Promises: Creating Promises, Consuming Promises, Promise Chaining, Promise.all. Async/Await Syntax. Async/Await Error Handling. Practical API Request Implementation.

MODULE III:

REACT FUNDAMENTALS

Introduction to React. Virtual DOM Concepts. JSX Syntax and Usage. React Developer Tools. Types of Components: Functional Components, Class Components, Props Communication.

Component Lifecycle Overview: Mounting Phase, Updating Phase, Unmounting Phase.

MODULE IV: REACT HOOKS, EVENT HANDLING

Hooks Overview, State Management using useState() hook. Side Effects Management using use Effect Hook, Cleanup Functions. React Hooks: use Hook, useReducer Hook usage and implementation. Creating Custom Hooks. Handling User Interactions. Event Handling Techniques. Conditional Rendering. Form Handling (Controlled and Uncontrolled Components). Dynamic Lists. Keys for Dynamic Content.React

MODULE V: ROUTING, CONTEXT AND AUTHENTICATION

Application Routing with React Router: Route Setup and Implementation, Navigation Components, Route Parameters, Nested Routes, useParams() hook, Advanced State Management: Context API for Global State, Reducer Patterns in React. Client-Server Communication. Authentication Flow with JWT: Protected Routes, Token Management. Security Best Practices.

LIST OF EXPERIMENTS:

1. Build a Quick Commerce Application (Zepto Clone)
 - a. Login/Logout Page
 - b. Products Page
 - c. Product Details Page
 - d. Cart Page
2. Build a Task Management Application(Trello Clone)
 - a. Login/Signup Page
 - b. Task Dashboard
 - c. Task Details Page
 - d. Add/Edit Task Page
3. Build a Emoji Game using Lists and keys
4. Build a Blog List by using React Router
5. Build a cowinDashboard Page using Third Party Packages
6. Get the popular github repos by using API Requests

REFERENCE BOOKS:

1. Front-End Web Development (UI/UX, HTML, CSS, JavaScript)
2. "JavaScript and JQuery: Interactive Front-End Web Development" by Jon Duckett
3. "Eloquent JavaScript" by Marijn Haverbeke

TEXT BOOKS:

1. "Learning React: Functional Web Development with React and Redux" by Alex Banks and Eve Porcello "CSS Secrets: Better Solutions to Everyday Web Design Problems" by Lea Verou

Employment Communication A Lab based course

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Course Objectives

After completing this course, students will be able to:

- Understand and apply principles of effective verbal and non-verbal communication to express ideas clearly and confidently in academic and professional contexts.
 - Demonstrate appropriate body language, listening skills, and cross-cultural communication practices to enhance interpersonal interactions.
 - Develop job-readiness skills by understanding the employment process and creating professional CVs and cover letters.
 - Participate effectively in Group Discussions (GDs) by articulating viewpoints logically, listening actively, and collaborating with team members.
 - Prepare for and perform successfully in personal interviews through structured responses, professional etiquette, and confidence-building practices.
 - Enhance overall communication competence and self-presentation skills through practical labs, peer interaction, and reflective activities.
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- **Week 1** : Course Introduction & Verbal Communication
 - **Week 2** : Non Verbal Communication & Cross Cultural Communication
 - **Week 3** : Body Language & Listening Skills
 - **Week 4** : The Employment Process & CV Writing
 - **Week 5** : The Cover Letter, CV Writing Lab & Group Discussions (GDs)
 - **Week 6** : Group Discussion (GD) Lab & Introduction to Interviewing
 - **Week 7** : Personal Interviews (PIs) & PI Practice Lab
 - **Week 8** : PIs, Student Speak, & Course Wrap-up

UNIVERSAL HUMAN VALUES - UNDERSTANDING HARMONY AND ETHICS

L	T	P	C
1	1	0	2

Course Outcomes

- Explain the need, basic guidelines, content and process of value education.
- Describe the harmony in the self and apply self-exploration as a process for inner development.
- Analyze relationships in family and society with clarity, trust and respect.
- Evaluate the interrelationship of individual, society and nature leading to sustainable living.
- Apply ethical principles in real-life personal, social, and professional situations.

MODULE- I

Introduction to Value Education and Harmony in Human Being, Need, Basic Guidelines, and Content of Value Education, Natural Acceptance and Experiential Validation, Understanding the Human Being as Co-existence of Self and Body, Harmony in the Self – Understanding Self (I)

MODULE- II

Harmony in the Family and Society, Understanding Values in Human Relationships, Trust, Respect, and Other Universal Values, Difference between Intention and Competence, Harmony in the Society – Universal Human Order

MODULE- III

Harmony with Nature and Existence, Understanding the Interconnectedness of Nature, Four Orders of Nature – Material, Plant/Bio, Animal, and Human, Coexistence of Mutual Fulfillment – Self and Body, Family and Society, Society and Nature, Living in Harmony with Nature – Eco-friendly Behavior and Sustainable Development

MODULE – IV

Professional Ethics and Social Responsibility, Definition and Importance of Professional Ethics, Competence in Professional Ethics, Integrity, Responsibility, and Ethical Human Conduct, Ethical Dilemmas and Case Studies

MODULE- V

Holistic Living and Universal Human Values, Vision for a Holistic Life and Universal Human Order, Realization and Fulfillment of Human Aspirations, Role of Education, Technology, and Mass Media in Value Education, Practicing Values – Real-life Applications and Activities

REFERENCES BOOKS:

1. "A Foundation Course in Human Values and Professional Ethics" Gaur, R. Sangal, G.P. Bagaria
2. "Human Values" A.N. Tripathi, "Professional Ethics and Human Values" Author: M. Govindarajan, S. Natarajan, V.S. Senthilkumar,
3. "The Bhagavad Gita" (for foundational ethical concepts and human values)
4. "The Dhammapada" (Buddhist text — values of compassion, right conduct, etc.)
5. "Ethics for the New Millennium" by Dalai Lama

FOREIGN LANGUAGE - II

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Course Objectives :

- To introduce learners to basic French language skills with emphasis on listening, speaking, reading, and writing.
- To enable students to communicate effectively in simple everyday situations using basic French expressions.
- To develop a foundational understanding of French grammar, vocabulary, and sentence structure.
- To help learners acquire correct pronunciation and intonation through systematic practice of sounds and accents.
- To familiarize students with French culture, social etiquette, and everyday practices to promote intercultural awareness.

MODULE I: Introduction to French

Importance of French as an international language, French-speaking countries, French alphabet and pronunciation, Greetings and leave-taking expressions, Self-introduction

MODULE II: Basic Grammar

Numbers (1–10), Days, months, time expressions, Family members, Professions.

MODULE III: Vocabulary and Expressions

Gender and plural forms of nouns, Personal pronouns

MODULE IV: Sentence Formation and Communication

Talking about daily activities, Describing people, places, and objects

MODULE V: Functional French and Culture

At the market and shopping, At a café or restaurant, Asking for directions, Travel and basic communication.